6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer:(penalty regime: 0 %)

Feedback

```
#include <stdio.h>


int main()
{
   int n, k;


   scanf("%d", &n);


   int a[n];


   for (int i = 0; i < n; i++)
   {
      scanf("%d", &a[i]);
   }


   scanf("%d", &k);
```

```
    int i = 0;
    int j = 1;

    while (i < n && j < n)
    {
        if (i == j)
        {
            j++;
        }
        else if (a[j] - a[i] == k)
        {
            printf("1");
            return 0;
        }
        else if (a[j] - a[i] < k)
        {
            j++;
        }
        else
        {
            i++;
        }
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
	3 1 3 5 4	1	1	
	10 1 4 6 8 12 14 15 20 21 25 1	1	1	
	10 1 2 3 5 11 14 16 24 28 29 0	0	0	
	10 0 2 3 7 13 14 15 20 24 25 10	1	1	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.